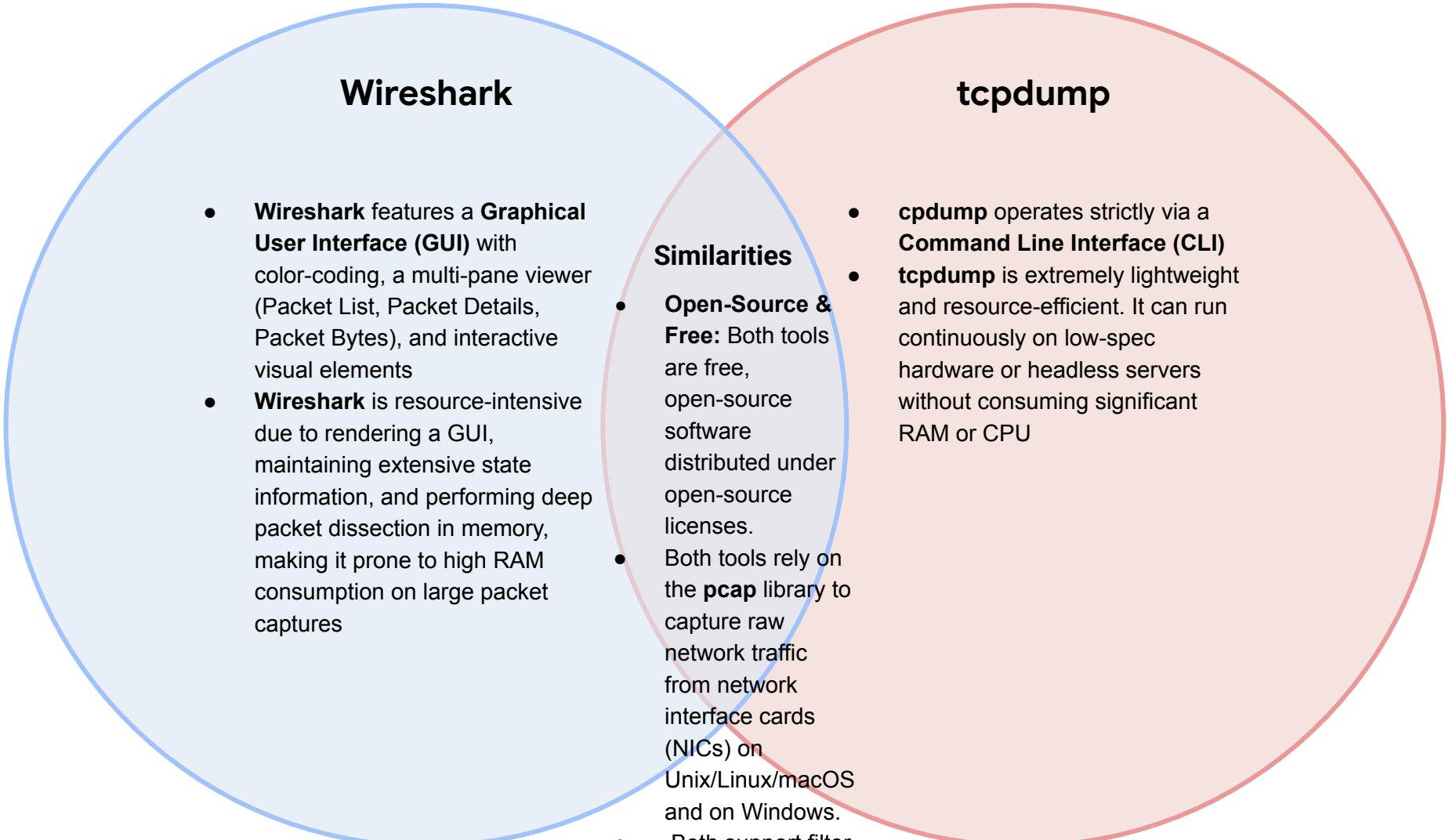

Wireshark
tcpdump
Similarities
• Wireshark features a Graphical User Interface (GUI) with color-coding, a multi-pane viewer (Packet List, Packet Details, Packet Bytes), and interactive visual elements
• Wireshark is resource-intensive due to rendering a GUI, maintaining extensive state information, and performing deep packet dissection in memory, making it prone to high RAM consumption on large packet captures
• Open-Source & Free: Both tools are free, open-source software distributed under open-source licenses.
• Both tools rely on the pcap library to capture raw network traffic from network interface cards (NICs) on Unix/Linux/macOS and on Windows.
• cpdump operates strictly via a Command Line Interface (CLI)
• tcpdump is extremely lightweight and resource-efficient. It can run continuously on low-spec hardware or headless servers without consuming significant RAM or CPU